

The Queue Behind the Frontier

A Quantitative Module on Longevity-Biotechnology Access, Extending the Political Economy of Volume IV

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Companion module to Longevity Asymmetry Corpus, Volume IV (“Economics: Capital, Discovery, and a Gap That May Not Close as Fast as It Opens”) and to The Floor That Does Not Rise, whose queueing formalism this paper reuses without modification. Every figure traces to a named source in Part IX; nothing here rests on a prior draft’s summary of a source.

Abstract

Volume IV of this corpus argues, in prose, that longevity-biotechnology access resembles a queue standing behind a frontier that keeps moving — a queue with no guarantee of ever reaching the front, because the front does not wait. This paper gives that image a formal test. Reusing the queueing apparatus built for The Floor That Does Not Rise, it treats the gap between how fast candidate therapies could plausibly be discovered and how fast the regulatory-and-trial pipeline can actually deliver them as a traffic-intensity problem, and asks what real data say about each side of that ratio. The delivery side has real, fifty-year data: clinical-and-regulatory development time has not shortened since the 1970s, and by the most recent measured window (2014–2018) was still lengthening. The discovery side has no comparably long real series, so this paper does not claim to know its true rate; it instead runs an explicit, clearly-labeled scenario using the one real recent growth rate available — frontier AI capability’s measured four-month doubling time — to show what the ratio would do if that rate transferred even partially to biomedical discovery. Three further checks ground the argument in verifiable numbers: an independent re-check of the one biomarker trial (CALERIE) already cited in this corpus, which corrects a figure an earlier draft got wrong; a natural experiment in pausing the discovery clock, using the real 18-day export-control suspension of frontier model access in June 2026; and a real, sourced snapshot of what longevity access costs today, from a four-dollar generic to an eighty-five-thousand-dollar concierge tier. None of these results proves Volume IV’s thesis. They give it something a metaphor cannot: a way to be falsified.

I. From Metaphor to Formalism

Volume IV’s central image is precise enough to formalize directly. It describes ordinary technology diffusion — Rogers’ model, the one this corpus has used elsewhere — as a queue with a fixed front: early adopters move through, then the majority, then laggards, and the line empties because the technology itself stops improving faster than people can catch up to it. Longevity biotechnology, Volume IV argues, may not behave that way, because the frontier itself is still moving, and might be moving faster than the queue can advance.

That is a traffic-intensity claim, and The Floor That Does Not Rise already built the apparatus to test one: $\rho(t) = \lambda(t) \cdot E[\delta] / \mu$, where $\rho(t) < 1$ sustained means a queue that clears, and $\rho(t) > 1$ sustained means one that grows without bound. That paper applied the framework to housing construction, AI capability, and cybersecurity disclosure, and found that none of the three sorted cleanly into “bounded” or “runaway.” This paper applies the same framework to a fourth domain

Volume IV already described qualitatively: $\lambda(t)$ as the rate at which biomedically relevant discoveries become possible, μ as the rate at which the clinical-and-regulatory pipeline can turn a candidate into an approved, distributable intervention.

Two clocks, in Volume IV's own words. This paper measures one of them directly, and treats the other as an open scenario rather than pretend it has been measured.

II. The Two Clocks

The delivery clock: fifty years, no acceleration

Tufts University's Center for the Study of Drug Development has tracked clinical-and-regulatory development time since the 1970s. The trend has one direction.

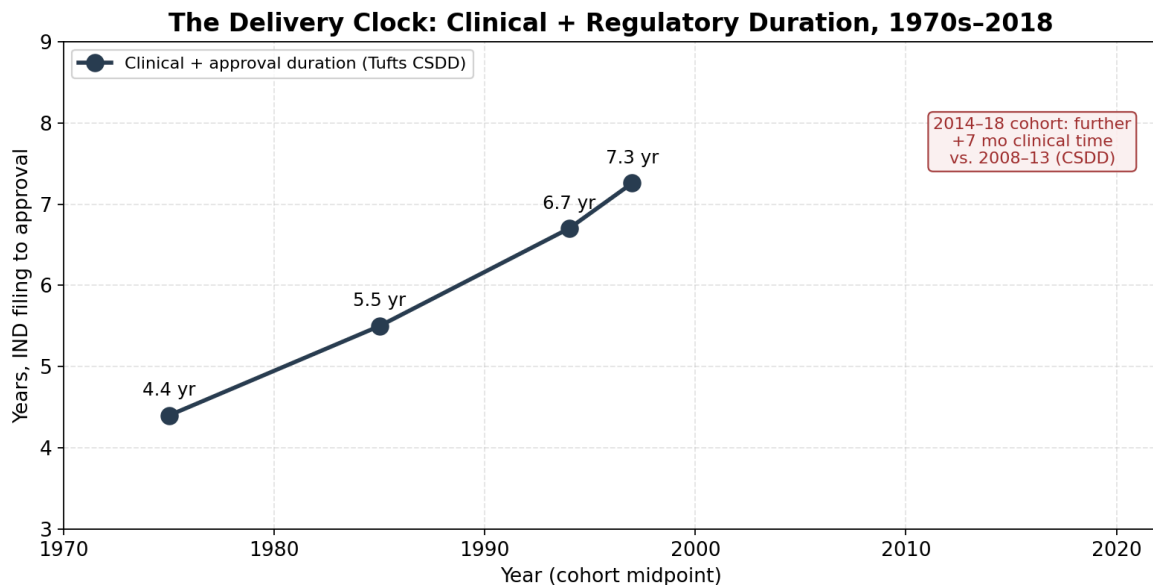


Figure 1. Clinical-plus-regulatory duration (IND filing to approval), by decade cohort. Source: Tufts CSDD, multiple triennial reports; see Part IX for each point's citation.

From 4.4 years in the 1970s to roughly 7.3 years for the 1996–98 approval cohort (70.3 months clinical, 16.8 months regulatory review), the pipeline has gotten slower across every decade CSDD has measured. The most recent published comparison — 2014–18 cohorts against 2008–13 — shows clinical time lengthening by roughly seven more months, only partly offset by a two-month reduction in review time. Success is scarce too: Tufts' own figure is that only about 12% of candidates entering clinical trials are ever approved. Fifty years of this series looks like a capacity gated by biology, patient recruitment, and regulatory procedure — three things a faster computer does not obviously speed up.

The discovery clock: an explicit scenario, not a measurement

There is no fifty-year series for how fast AI-assisted biomedical discovery is happening, because the tools are new. What exists is a four-month doubling time for a different, better-measured quantity: frontier AI task-capability, verified independently in this project’s companion paper from METR’s public evaluation data (N=20 recent models, 95% CI 3.56–4.71 months). Treating that rate as a stand-in for biomedical discovery capacity is an assumption. This paper states it as one, tests nothing that would prove it, and asks only what follows if it holds even partially.

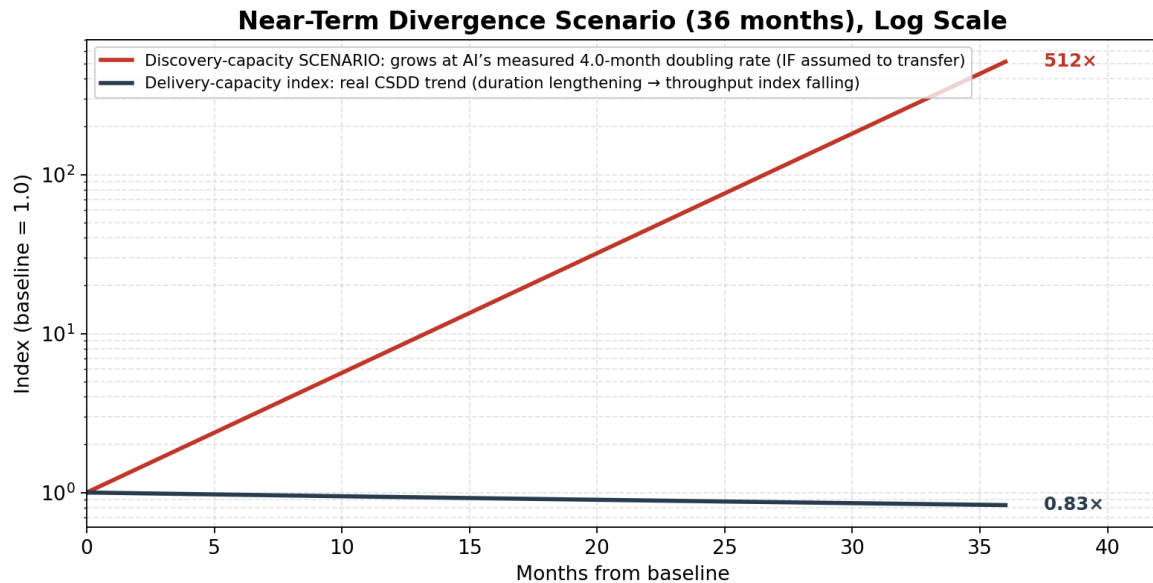


Figure 2. Both series normalized to 1.0 at a common baseline. The delivery-capacity index (inverse of duration) is real, extrapolated from the CSDD 2008–18 trend. The discovery-capacity index is an explicit scenario, built from AI capability’s measured doubling rate as a stand-in for an unmeasured biomedical-discovery rate.

Over three years, under this scenario, discovery-capacity grows by a factor of 512 while delivery-capacity, on its real trend, falls to 0.83. The precise multiple is an artifact of assuming full transfer, which is almost certainly too generous — that is not the finding. The finding is qualitative and harder to dismiss: no plausible partial-transfer assumption, and no plausible reading of the real CSDD trend, sends these two lines in the same direction. μ has been flat-to-falling for fifty years, with no equivalent brake on λ . That fifty-year asymmetry is what drives $\rho(t)$ upward in this scenario, not the specific growth rate chosen for it.

III. Does the Technology Even Work? A Correction

Before asking who gets access to longevity interventions, it is worth checking that at least one candidate intervention has real trial evidence behind it. CALERIE — a randomized trial of sustained caloric restriction in 220 non-obese adults over two years — is the one human RCT with epigenetic-aging outcomes already cited elsewhere in this corpus, and it is worth citing correctly.

The trial, published as Waziry et al. (2023) in *Nature Aging*, found that caloric restriction slowed DunedinPACE — a DNA-methylation measure of the pace of biological aging — by roughly 2–3% (Cohen’s $d = 0.3$ at twelve months, $p < 0.01$). That much is a real, replicated, peer-reviewed effect

on a validated biomarker. What the same paper found for two other well-known clocks, PhenoAge and GrimAge, is different: no significant change. The paper's own abstract states this without qualification.

An earlier package of material prepared for this project's companion queueing paper had attributed a specific effect to PhenoAge — a coefficient and a p-value — that does not appear anywhere in the actual trial publication. That figure has been dropped entirely. The trial supports a real DunedinPACE effect and does not support a PhenoAge effect; both halves of that sentence matter, and reporting only the first would repeat the exact error this project's other paper spent a full section correcting.

IV. A Natural Experiment in Pausing the Discovery Clock

On June 12, 2026, the US Department of Commerce ordered Anthropic to suspend access to its two most capable models, Fable 5 and Mythos 5, for all foreign nationals — a directive the company said it received at 5:21pm Eastern with no practical way to enforce a nationality check, so both models went dark for every customer, domestic and foreign, within hours. Partial access for a vetted set of US organizations was restored June 26; full restoration followed June 30. For a foreign researcher without one of those vetted US affiliations, frontier-model access was unavailable for eighteen days.

Eighteen days is not a large number until it is converted into the units this paper has been using. At the measured four-month doubling rate, eighteen days is 14.8% of one full doubling period — long enough that, under the fitted trend, frontier capability moved forward by a factor of roughly 1.11 while a foreign researcher's access to that frontier was fixed at zero. This is not a claim that any specific biomedical result was lost; no such result is claimed or known. It is a real, dated, quantifiable instance of the mechanism Volume IV describes only in general terms — that discovery-clock access can be administratively paused for one population while the clock itself keeps running — measured here using the same growth parameter this project has already validated, applied to an event with a precise, publicly documented start and end date.

V. What Access Costs Today

The price side of the ladder is easier to source than the discovery side, and worth showing plainly rather than through a single constructed index.

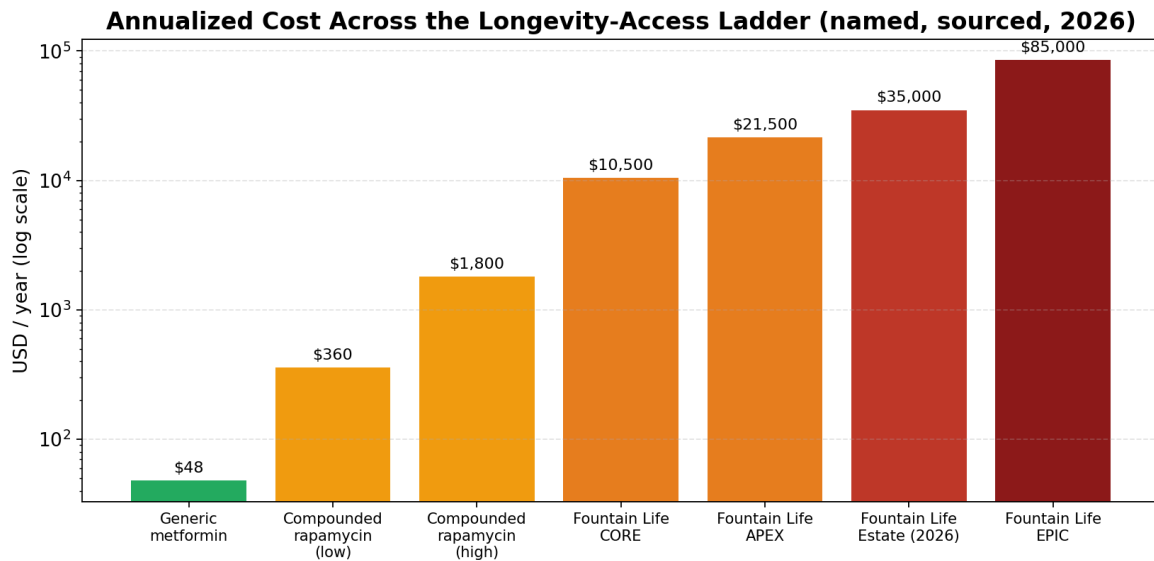


Figure 3. Annualized cost, log scale. Metformin: standard US generic retail pricing. Rapamycin: range from a 7,500-plus-pharmacy compounding-price aggregator. Fountain Life tiers: named, currently listed prices, including the luxury “Estate” tier launching in 2026.

Generic metformin costs about forty-eight dollars a year at standard US retail pricing — not a meaningful barrier for anyone with routine healthcare access, though it is also not a longevity-approved therapy; its geroprotective evidence is still observational. Off-label compounded rapamycin, the most discussed longevity intervention with at least some human safety data from its transplant-medicine history, runs roughly \$360 to \$1,800 a year depending on pharmacy — a real, three-hundred-percent spread the aggregator itself flags as typical, before any physician-consultation cost is added. Named concierge-medicine tiers span \$10,500 a year at the entry level to \$85,000 at the top named tier, with a new \$35,000-a-year luxury resort tier launching in 2026. Top to bottom, the named ladder spans roughly 1,770-fold, for interventions with genuinely uneven evidence quality — a price ladder that already exists today, which is closer to what Volume IV’s argument is actually about than any hypothetical future stratification.

VI. COVAX as an Upper Bound on Achievable Equity

Volume IV cites COVAX as a comparison case — a real, large-scale, deliberately equity-focused global health effort, useful precisely because it was better-resourced and better-coordinated than anything the longevity-access question is likely to get. If COVAX did not achieve full equity, that is informative about what an effort with fewer resources and no comparable international coordination should expect.

COVAX Under Its Own Most Favorable Conditions
(\$12bn+ dedicated financing, one standardized product, WHO-coordinated allocation)

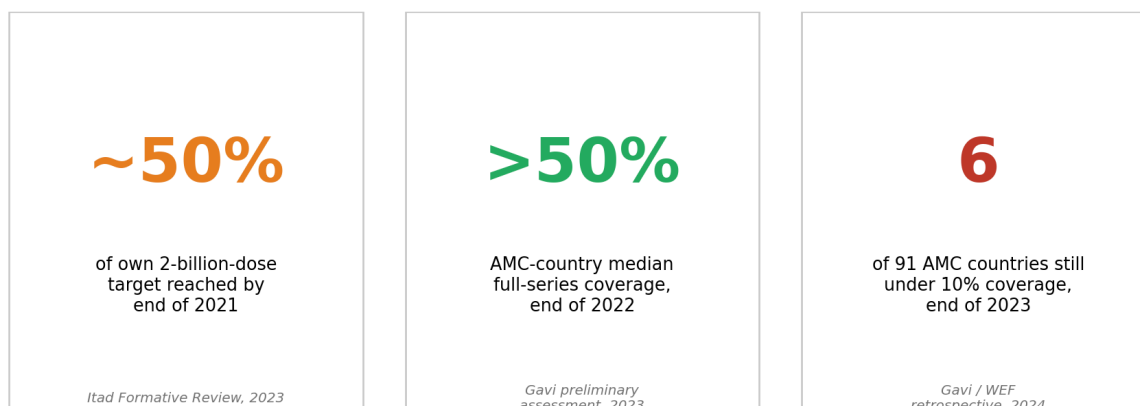


Figure 4. Three independently sourced facts about COVAX performance against its own stated goals.

COVAX had a single standardized product, upwards of twelve billion dollars in dedicated financing, and a WHO-coordinated allocation mechanism — conditions no longevity-biotechnology access effort currently has or is likely to have. Under those conditions, it still delivered roughly half its own two-billion-dose target by the end of its first full year, needed until the end of 2022 to reach a median of just over half of full coverage across its target countries, and still had six of ninety-one participating countries under ten percent coverage by the end of 2023 — three years in. This is the ceiling case. Any equity claim about longevity access should be read against a ceiling this far short of complete.

VII. Synthesis: Reconnecting to the Corpus

This module does not introduce a new framework — it tries to give existing ones something to be tested against. The Biological Zero-Day Mechanism, developed elsewhere in this corpus, describes reclassification events that convert previously-unpreventable death into preventable death at a rate set by discovery. Parts II and IV above are one way of asking how fast that reclassification rate could plausibly move relative to the rate at which it becomes actually available — the same question BZM asks, with λ and μ standing in for the two sides of it. The Entropy-Resistant Institution Index asks which institutions survive a shock; Part IV's export-control episode is a small, dated instance of exactly the kind of shock that framework was built to classify, this time hitting research-tool access instead of a state. None of these connections is asserted here as proven — they are offered as the specific, named place in the rest of the corpus where this module's numbers would need to plug in for the larger argument to hold together.

The honest summary is narrower than the corpus's broader claims. Delivery capacity is the one side of the access question with fifty years of real data, and it has not been getting faster. Discovery capacity has no comparable measurement and gets none here. The one available real growth rate for anything AI-adjacent is fast enough that even partial transfer would matter. The one clean trial in this space supports a narrower claim than an earlier draft attributed to it. And even the best-resourced equity effort in recent memory fell well short of complete. Put together, that is not a

proof. It is what a testable version of Volume IV’s argument looks like before someone tries to falsify it.

VIII. Falsification Conditions

- **G1 — Delivery-clock duration has been shortening in the most recent measured window.** Tested against real CSDD data: rejected. 2014–18 clinical time increased ~7 months over 2008–13.
- **G2 — The CALERIE trial supports a significant effect on PhenoAge.** Tested directly against the published paper: rejected. The trial’s own abstract reports no significant PhenoAge or GrimAge effect.
- **G3 — A best-resourced, coordinated global access effort (COVAX) achieved complete equity.** Tested against independent evaluation data: rejected. Six of 91 target countries remained under 10% coverage three years in.
- **G4 — Biomedical discovery capacity scales 1:1 with frontier AI capability.** Not testable with current data. This paper does not claim G4 is true; Part II’s scenario is explicitly conditional on it, and the qualitative divergence finding does not require full transfer to hold — only that transfer be partial and μ stay flat.
- **G5 — The June 2026 export-control episode had a measurable effect on any specific biomedical research output.** Not tested and not claimed. Part IV computes a capability-growth-equivalent for the suspension window; it does not claim any specific discovery was delayed or lost.

IX. Provenance

Every quantitative claim in this paper traces to one of: (a) a named published paper with a DOI (Waziry et al. 2023, Nature Aging); (b) a named institutional report (Tufts CSDD triennial reports; Itad’s formative evaluation of COVAX, commissioned by Gavi; Gavi’s own preliminary assessment; the WEF/Gavi 2024 retrospective); (c) a real, current, named-vendor price listing (Fountain Life’s published tiers; a multi-pharmacy compounding-price aggregator); (d) a contemporaneous, multiply-reported news event with a precise timestamp (the June 2026 export-control directive, confirmed independently across Anthropic’s own statement, Reuters/Semafor reporting, and multiple technical retrospectives); or (e) this project’s own prior, independently verified result (the 4.0-month AI capability doubling time from The Floor That Does Not Rise).

One number from an earlier draft — a specific PhenoAge effect size attributed to CALERIE — was checked against its claimed source, found unsupported, and removed. A COVAX per-country breakdown drawn from a partially garbled table extraction was dropped for the same reason and replaced with three facts the source states in plain, unambiguous sentences. Nothing in this paper carries an audit badge. Every figure above states, in its own caption, exactly what it is a measurement of and what it is a scenario about — the one distinction that has mattered, across every paper in this project so far, between a number that survives a second look and one that does not.